

Mohamed Aziz Chouaibi

Embedded Systems & Computer Vision — IIA Engineering Student

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EDUCATION

- National Institute of Applied Science and Technology (INSAT)** Tunis, Tunisia
 - Engineering Degree — Industrial Computing and Automation (IIA)* Sep 2022 – Present
 - Coursework focus: embedded systems, control, computer vision, applied AI.
 - Hands-on projects with microcontrollers, Jetson Nano, and PLCs.

CERTIFICATIONS

- Deep Learning Track — DataCamp** 2023
 - Credential*
- Supervised Machine Learning: Regression and Classification — Coursera** 2023
 - Credential*

EXPERIENCE

- Computer Vision Developer** GENOW — HawkVision Project
 - Part-time* Sep 2025 – Present
 - Advancing the HawkVision embedded AI pipeline by optimizing object detection and segmentation modules for real-time Jetson Nano deployment using ONNX.
- Computer Vision Developer — Intern** GENOW — HawkVision Project
 - Hybrid* Jul 2025 – Aug 2025
 - Developed and optimized computer vision models for the HawkVision ADAS system.
- Embedded Systems Developer** Research Collaboration — Shaqra University (Saudi Arabia)
 - Research with Prof. Rim Dhaou Hamdaoui* 2024
 - Designed an RFID-based solution for intelligent crowd management at Al-Rawdah Al-Sharifah.
 - Prototype developer for a peer-reviewed publication in the Journal of Engineering Sciences and Information Technology.

PROJECTS

- Deep Learning for Retinal Disease Detection** Personal Project
 - 2025 – Ongoing*
 - Built a PyTorch classifier for diabetic retinopathy stages.
 - Applied OpenCV preprocessing (CLAHE, cropping, green-channel).
- Smart Factory Automation (TIA Portal & Factory I/O)** Academic Project
 - 2025*
 - Programmed Siemens S7-1500 PLCs to operate a virtual production line.
- CH-Blocker: Python Website Blocking Software** Independent Project
 - 2021*
 - Windows/Linux app modifying hosts file with scheduling UI.

SKILLS SUMMARY

- Languages:** Python, C++, Bash
- Machine Learning / Deep Learning:** Machine Learning, PyTorch, YOLO, ONNX, OpenCV
- Embedded / Edge:** Jetson Nano, ESP32, RTOS basics
- Automation:** TIA Portal, PLC (Ladder/FB)
- Tools:** Linux, Gits
- Focus Areas:** Computer Vision, Embedded AI, Industrial Automation

EXTRACURRICULAR ACTIVITIES

- Logistics Member — Android INSAT Club**
 - 2025*
 - Supported event organization, logistics, and workshop operations.